KAUSTUBH DATTATRAY SHINDE

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PROFESSIONAL SUMMARY

Electronics & Telecommunication Engineer with hands-on development experience in Java, React and SQL, alongside applied AI and embedded-systems project work. Built an AI-powered autonomous vehicle combining computer vision, NLP and real-time control. Comfortable working across the stack and quickly picking up new frameworks and tools to solve real problems.

EDUCATION

ISBM College of Engineering, Pune 2022 – 2026
Bachelor of Engineering — Electronics & Telecommunication Engineering, Savitribai Phule Pune University (SPPU)
CGPA: FY 7.59 | SY 9.11 | TY 8.80 | BE 9.18

RESEARCH PUBLICATIONS

Advanced Safety System for Two-Wheeler Using PIC16F877A and CAN Bus ICETT-2026
International Conference on Emerging Trends in Technology — Co-author
- Co-authored a research paper proposing a low-cost embedded safety system for two-wheelers built around a PIC16F877A microcontroller with CAN bus communication for reliable, fault-tolerant data exchange.
- Contributed to the system architecture covering impact detection, CO gas monitoring and PT100-based engine temperature sensing, with GSM-based emergency alerting.

AI-Powered Autonomous RC Vehicle: Design and Performance Final Year Project Report
SPPU Stage-I Report — Author
- Authored the Phase-I project report covering literature survey, problem formulation, system architecture and proposed methodology for an AI-powered autonomous RC vehicle.
- Detailed the integration of voice-command recognition, conversational AI processing, motor control and computer-vision-based obstacle detection into a single Raspberry Pi platform.

PROJECTS

GameShop | Gaming Café Booking System
Java 21, Spring Boot, Spring Security, React, MySQL, JDBC
- Developed a full-stack gaming café booking system with a React customer booking interface and an administrative dashboard for managing bookings and gaming stations.
- Implemented REST endpoints for creating, retrieving, confirming and cancelling bookings, with validation for booking rules and station availability.
- Built station-assignment logic to allocate an available operational station when an admin confirms a booking, and release the assignment when a booking is cancelled.
- Integrated MySQL through JDBC and configured Spring Security for session-based admin authentication, role-based authorization and CSRF protection.

AI-Powered Autonomous RC Car | Voice Control & Navigation
Raspberry Pi, Python, Flask, OpenCV, Gemini API, GPIO
- Built an autonomous RC vehicle on a Raspberry Pi 4 with a Flask backend, live OpenCV video streaming, and a Gemini vision-model agent that analyses the camera feed to make real-time driving decisions.
- Implemented a multithreaded GPIO motor-control module for the drive system, with an independent safety loop to manage obstacle response and emergency stop.
- Developed a voice-command pipeline that routes transcribed natural-language instructions through a Gemini-powered agent to executable drive commands (forward, backward, left, right, stop).
- Built a browser-based control dashboard serving a live MJPEG camera stream alongside manual drive controls and an autopilot toggle.

Amazon Clone | E-Commerce UI
JavaScript (ES6 Modules), HTML5, CSS3, Jasmine
- Built a responsive Amazon-style storefront in vanilla JavaScript, HTML and CSS with a cart, multi-step checkout, delivery-option selection and an order-tracking view, persisting cart state via local storage.
- Refactored the cart module into a class-based (OOP) design and wrote unit tests for cart and currency-formatting utilities using the Jasmine testing framework.

TECHNICAL SKILLS

Languages: Java, JavaScript, SQL, HTML5, CSS3
Frameworks: React, Spring Boot
Database: MySQL
Developer Tools: Git, GitHub, VS Code
Concepts: Object-Oriented Programming
Certifications: Java Programming Masterclass (Udemy), React Developer (HackerRank), SQL Advanced (HackerRank)